

diode and resistor can make a simple and inexpensive voltage regulator. Be sure to select the proper wattage for the zener and the proper wattage and resistance for the resistor. Zener diodes are also rated by their tolerance (1 percent and 5 percent are common) and their power rating, in watts. Note the resistor in the schematic shown in the figure given alongside.

This resistor (R) limits the current through the zener, and its value (and wattage) is determined by the current draw from the load, as well as the input and output voltages. The process of determining the correct values and ratings for resistor R and the zener diode is fairly simple and uses the basic electricity rules. The zener voltage rating is, quite obviously, the desired regulated voltage—you may find that the available rated voltages are somewhat awkward (such as 5.1 V), but you should be able to find a value within a few percent of the rated value. Once you know the voltage rating for the zener diode that you are going to use, you can then calculate the value and ratings for R. The zener diode regulator shown in the figure alongside is actually a voltage divider, with the lower portion being a set voltage level. To determine the correct resistance of R, you have to know what the input voltage is and the current that is going to be drawn from the regulator. For example, if you wanted 100 mA at 5.1 V from a 12V power supply, the resistance of R can be calculated as:

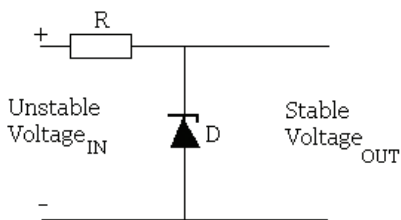
$$V_{I\max} = 12 - 5.1 = 6.9 \text{ V}$$

$$I_{\max} = \frac{V}{R} = \frac{6.9}{100} = 0.069 \text{ A}$$

The closest “standard” resistor value you can get is 68 ohms, which will result in 101 mA being available for the load. With this value in hand, you can now calculate the power being dissipated by the resistor, using the basic power formulas:

$$P = V \times I = 6.9 \times 0.101 = 0.6969 \text{ W}$$

Standard resistor power ratings (in watts) are 1/8, 1/4, 1/2, 1, 2, and so on. A 1-W, 68 ohm resistor would be chosen for this application. The zener diode will also be dissipating power. The voltage drop here is something that you



A simple Zener diode voltage regulator

need to decide. To be on the safe side, it is recommended that you assume that the zener diode can have 100 percent of the load current passing through it (when the load circuitry is not attached to the power supply). The power rating for the zener diode is calculated exactly the same way as R: